

Math 340
Practice Final Exam
Time Limit: 120 Minutes

Name (Print): _____
TA's name and Section#: _____
Student ID Number: _____

This exam contains 10 pages (including this cover page) and 12 problems. The last page is the scratch paper. If any pages are missing, please inform the instructor immediately.

Directions:

- Enter all requested information on the top of this page, and put your initials on the top of every page, in case the pages become separated.
- You may *not* use your calculators, books and notes.
- You must show all your work. A stranger should be able to pick up your paper and follow your reasoning. You must use proper notation and define the events clearly. Partial credits may be given.
- Unless otherwise noted, your answers must be in fraction form or terminating decimals.
- Circle your answer to the True/False problems and justify your answer.
- For the free response problems you need to show all necessary work in the space provided for each problem. Answers must be justified with sufficient work, and partial credit will be given for appropriate work shown.

I do not lie, cheat or steal, or tolerate those who do.

By signing below you indicate that all your work is your own and that you have neither given nor received help from external sources.

Signature: Solutions

Total: _____

T/F		P. 9	
P.6		P. 10	
P.7		P. 11	
P.8		P. 12	

True/False Questions: If true, give your justification. If false, give a counterexample or some reasoning.

1. (2 points) Let V be a vector space of dimension 4. If $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$ is a basis of V , then so is $\{\mathbf{v}_1 + \mathbf{v}_2, \mathbf{v}_2 + \mathbf{v}_3, \mathbf{v}_3 + \mathbf{v}_4, \mathbf{v}_4 + \mathbf{v}_1\}$.

A. True.

☒ B. False.

This is not a linearly independent set since
 $(\vec{v}_1 + \vec{v}_2) + (-1)(\vec{v}_2 + \vec{v}_3) + (\vec{v}_3 + \vec{v}_4) + (-1)(\vec{v}_4 + \vec{v}_1) = \vec{0}$
(If you can't see this directly, set up $a_1(\vec{v}_1 + \vec{v}_2) + a_2(\vec{v}_2 + \vec{v}_3) + a_3(\vec{v}_3 + \vec{v}_4) + a_4(\vec{v}_4 + \vec{v}_1) = \vec{0}$ and solve for a_1, a_2, a_3, a_4 . You'll find there are nontrivial solutions.)

2. (2 points) If $\mathbf{v}$ is orthogonal to both $\mathbf{u}$ and $\mathbf{w}$, then $\mathbf{v}$ is orthogonal to the space spanned by $\{\mathbf{u}, \mathbf{w}\}$.

☒ A. True.

B. False.

let $\vec{x} \in \text{span}\{\vec{u}, \vec{w}\}$,

every vector in

then $\vec{x} = a\vec{u} + b\vec{w}$ for some $a, b \in \mathbb{R}$

$$\begin{aligned} \text{Then } \vec{v} \cdot \vec{x} &= \vec{v} \cdot (a\vec{u} + b\vec{w}) = \vec{v} \cdot (a\vec{u}) + \vec{v} \cdot (b\vec{w}) \\ &= a\vec{v} \cdot \vec{u} + b\vec{v} \cdot \vec{w} = 0 \quad (\text{since } \vec{v} \cdot \vec{u} = 0 = \vec{v} \cdot \vec{w}) \end{aligned}$$

3. (2 points) Let A be an $n \times n$ matrix. If A is singular, then 0 is an eigenvalue of A .

☒ A. True.

B. False.

*Since A is singular, $A\vec{x} = \vec{0}$ has a nonzero solution,
 That is, $A\vec{x} = \vec{0} = 0\vec{x}$ for some $\vec{x} \neq \vec{0}$.*

This shows 0 is an eigenvalue of A .

- ~~4.~~ (2 points) Let A be any square matrix. Then the matrix $A + A^T$ is always diagonalizable.

A. True.

B. False.

- ~~5.~~ (2 points) All eigenvalues of a symmetric real matrix are real and positive.

A. True.

B. False.

6. (10 points) Let

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & -2 \end{bmatrix} \text{ and } B^{-1} = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 1 \\ 1 & 2 & 4 \end{bmatrix}.$$

(a) Compute $\det(A)$.

$$\begin{aligned} \det A &= 1 \cdot \det \begin{bmatrix} 0 & 1 \\ 1 & -2 \end{bmatrix} - 1 \cdot \det \begin{bmatrix} 2 & 0 \\ 1 & -2 \end{bmatrix} \\ &= 1 \cdot (0 - 1) - 1 \cdot (-4 - 0) \\ &= -1 + 4 = 3 \end{aligned}$$

(b) Compute $\det(B)$.

First let's find $\det(B^{-1})$. You can compute it by cofactor expansion like above or do:

$$\begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 1 \\ 1 & 2 & 4 \end{bmatrix} \xrightarrow[\text{doesn't change det.}]{\substack{-3r_1 + r_2 \rightarrow r_2 \\ -r_1 + r_3 \rightarrow r_3}} \begin{bmatrix} 1 & 2 & 3 \\ 0 & -5 & -8 \\ 0 & 0 & 1 \end{bmatrix}$$

B^{-1}

$$\text{so } \det(B^{-1}) = 1 \cdot (-5) \cdot 1 = -5.$$

$$\begin{aligned} \text{Then } \det(B) &= \frac{1}{\det(B^{-1})} \\ &= -\frac{1}{5} \end{aligned}$$

(c) Compute $(A^{-1}B^T)^{-1}$.

$$\begin{aligned} (A^{-1}B^T)^{-1} &= (B^T)^{-1}(A^{-1})^{-1} \\ &= (B^{-1})^T A \\ &= \begin{bmatrix} 1 & 3 & 1 \\ 2 & 1 & 2 \\ 3 & 1 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & -2 \end{bmatrix} \\ &= \begin{bmatrix} 4 & 3 & 1 \\ 3 & 6 & -3 \\ 4 & 10 & -7 \end{bmatrix} \end{aligned}$$

7. (10 points) Let $L: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be defined by

$$L \left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \right) = \begin{bmatrix} x_1 + x_3 \\ 2x_1 + x_2 + x_3 \\ 3x_1 + x_2 \end{bmatrix}.$$

(a) Find $\text{Ker} L$.

Let $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3$. Then

$\vec{x} \in \text{Ker } L$

$$\Leftrightarrow L(\vec{x}) = \vec{0}$$

$$\Leftrightarrow \begin{cases} x_1 + x_3 = 0 \\ 2x_1 + x_2 + x_3 = 0 \\ 3x_1 + x_2 = 0 \end{cases} \quad *$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 2 & 1 & 1 & 0 \\ 3 & 1 & 0 & 0 \end{array} \right] \xrightarrow{\substack{-2r_1 + r_2 \\ -3r_1 + r_3}} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 1 & -3 & 0 \end{array} \right] \xrightarrow{-r_2 + r_3} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & -2 & 0 \end{array} \right]$$

$$\xrightarrow{-\frac{1}{2}r_3} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right]$$

So the only solution to *

$$\text{is } \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{Thus } \text{Ker } L = \left\{ \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \right\}.$$

(b) Is L onto? Give your reason.

Yes.

By the rank-nullity theorem, $\dim(\text{Ker } L) + \dim(\text{range } L) = \dim(\text{domain of } L)$
 $= 3$

$$\text{So } \dim(\text{range } L) = 3 - \dim(\text{Ker } L) = 3 - 0 = 3$$

Thus $\text{range } L$ is a 3-dimensional subspace of $\mathbb{R}^3$ so has to be all of $\mathbb{R}^3$.

(c) Is L invertible? If so, find $L^{-1} \left(\begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} \right)$. (You can also show L is onto directly.)

By (a), (b), L is one-to-one and onto, so it is invertible and L^{-1} is

$$\text{defined. } L^{-1} \left(\begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} \right) = \begin{bmatrix} a \\ b \\ c \end{bmatrix} \Leftrightarrow L \left(\begin{bmatrix} a \\ b \\ c \end{bmatrix} \right) = L \left(L^{-1} \left(\begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} \right) \right) = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$$

$$\text{So } \begin{cases} a + c = 2 \\ 2a + b + c = 1 \\ 3a + b = 1 \end{cases} \rightarrow \begin{cases} a + c = 2 \\ b - c = -3 \\ -2c = -2 \end{cases} \rightarrow \begin{cases} a = 1 \\ b = -2 \\ c = 1 \end{cases}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 2 & 1 & 1 & 1 \\ 3 & 1 & 0 & 1 \end{array} \right] \xrightarrow{\substack{-2r_1 + r_2 \\ -3r_1 + r_3}} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & -1 & -3 \\ 0 & 1 & -3 & -5 \end{array} \right] \xrightarrow{-r_2 + r_3} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & -1 & -3 \\ 0 & 0 & -2 & -2 \end{array} \right]$$

$$\text{So } L^{-1} \left(\begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}.$$

You could also find the formula for L^{-1} first.

8. (10 points) Let A be the following matrix

$$A = \begin{bmatrix} 1 & 1 & -1 \\ 0 & 2 & 0 \\ -1 & 1 & 1 \end{bmatrix}.$$

(a) Compute the characteristic polynomial of A .

The char. poly of A is

$$\det(\lambda I - A) = \det \begin{bmatrix} \lambda-1 & -1 & 1 \\ 0 & \lambda-2 & 0 \\ 1 & -1 & \lambda-1 \end{bmatrix} = (\lambda-1)(\lambda-2)(\lambda-1) + 1(0-(\lambda-2))$$

$$= (\lambda-2)(\lambda-1)^2 - (\lambda-2) = (\lambda-2)((\lambda-1)^2 - 1) = (\lambda-2)(\lambda-2)\lambda = (\lambda-2)^2 \lambda$$

(It's also ok to write $\lambda^3 - 4\lambda^2 + 4\lambda$ as your final answer here. I factored it for the sake of part (b).)

(b) Determine the eigenvalues of A and their algebraic multiplicities.

(If you haven't factored in the above, factor here.)

eigenvalues: 2 and 0, where 2 has algebraic multiplicity 2 and 0 has algebraic multiplicity 1.

(c) Is A diagonalizable? If so, find a diagonalization (That is, find matrices D and P such that $A = PDP^{-1}$). If not, explain why.

Eigenspace assoc. w/ eigenvalue 2:

$$\text{Solve } (\lambda I - A) \vec{v} = \vec{0}$$

$$\text{i.e. } \begin{bmatrix} 1 & -1 & 1 \\ 0 & 0 & 0 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} t-s \\ t \\ s \end{bmatrix} = t \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + s \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \quad t, s \in \mathbb{R}.$$

Eigenspace assoc. w/ eigenvalue 0:

$$\text{Solve } (0I - A) \vec{v} = \vec{0}$$

$$\text{i.e. } \begin{bmatrix} -1 & -1 & 1 \\ 0 & -2 & 0 \\ 1 & -1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{cases} -x - y + z = 0 \\ -2y = 0 \\ x - y - z = 0 \end{cases} \Rightarrow \begin{cases} -x + z = 0 \\ y = 0 \\ x - z = 0 \end{cases} \Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} r \\ 0 \\ r \end{bmatrix} = r \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad r \in \mathbb{R}.$$

The choice of P is not unique.

For example you can scale any column by a nonzero scalar.

If you permute the columns of P , you need to permute diagonal entries of D accordingly. $P = [\vec{v}_1 \ \vec{v}_2 \ \vec{v}_3]$ and $D = \begin{bmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \lambda_3 \end{bmatrix}$ come in pairs where $A\vec{v}_i = \lambda_i \vec{v}_i$.

So $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ are l.i. eigenvectors w/ eigenvalues 2, 2, 0

respectively. Let $P = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}$, and $D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$, then $P^{-1}AP = D$ (i.e. $A = PDP^{-1}$)

9. (10 points) Find an orthonormal basis for the column space (i.e., the subspace spanned the column vectors) of the matrix:

Gram-Schmidt

$$A = \begin{bmatrix} 1 & -1 & 1 \\ 0 & 0 & 1 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \end{bmatrix}.$$

10. (10 points) Let P_2 be the space of polynomials with $\mathbb{R}$ -coefficients of degree ≤ 2 . We define an inner product on P_2 by

$$(f, g) = \int_0^1 f(t)g(t)dt.$$

- (a) Let $g_1 = t - 1$ and $g_2 = t^2$. Let θ be the angle between g_1 and g_2 . Compute the norms (or lengths) of g_1, g_2 and $\cos(\theta)$.

$$(g_1, g_1) = \int_0^1 g_1(t)g_1(t)dt = \int_0^1 (t-1)^2 dt = \int_0^1 t^2 - 2t + 1 dt = \left. \frac{1}{3}t^3 - t^2 + t \right|_0^1 = \frac{1}{3} - 1 + 1 = \frac{1}{3}$$

$$(g_2, g_2) = \int_0^1 g_2(t)g_2(t)dt = \int_0^1 t^4 dt = \left. \frac{1}{5}t^5 \right|_0^1 = \frac{1}{5}$$

$$(g_1, g_2) = \int_0^1 g_1(t)g_2(t)dt = \int_0^1 t^3 - t^2 dt = \left. \frac{1}{4}t^4 - \frac{1}{3}t^3 \right|_0^1 = \frac{1}{4} - \frac{1}{3} = -\frac{1}{12}$$

$$\text{So } \|g_1\| = \sqrt{(g_1, g_1)} = \frac{1}{\sqrt{3}}, \quad \|g_2\| = \sqrt{(g_2, g_2)} = \frac{1}{\sqrt{5}}, \quad \cos \theta = \frac{(g_1, g_2)}{\|g_1\| \|g_2\|} = -\frac{\sqrt{15}}{12}$$

- (b) Compute the closest distance between g_2 and vectors in the span of g_1 .

(Projection.)



closest distance

$$= \|g_2\| \sin \theta$$

$$= \frac{1}{\sqrt{5}} \sqrt{1 - \frac{15}{144}}$$

$$= \frac{\sqrt{129}}{12\sqrt{5}}$$

11. (10 points) Let $L: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be defined by

$$L\left(\begin{bmatrix} u_1 \\ u_2 \end{bmatrix}\right) = \begin{bmatrix} 2u_1 + u_2 \\ u_1 + 3u_2 \end{bmatrix}.$$

Let $S = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \end{bmatrix} \right\}$ be the ordered basis for $\mathbb{R}^2$. For this question, evaluate all matrix inverses and products

(a) Find the matrix representation of L with respect to the natural basis $E = \{\vec{e}_1, \vec{e}_2\}$ for $\mathbb{R}^2$.

$$L\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \quad \text{so} \quad [L(\begin{bmatrix} 1 \\ 0 \end{bmatrix})]_E = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$L\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \quad \text{so} \quad [L(\begin{bmatrix} 0 \\ 1 \end{bmatrix})]_E = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

Thus the matrix representation of L w.r.t. E is

$$\begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}. \quad (\text{Can also observe } L\left(\begin{bmatrix} u_1 \\ u_2 \end{bmatrix}\right) = \begin{bmatrix} 2u_1 + u_2 \\ u_1 + 3u_2 \end{bmatrix})$$

(b) What is the transition matrix $P_{S \leftarrow E}$ from the E -basis to the S -basis?

$$P_{S \leftarrow E} = \begin{bmatrix} [\vec{e}_1]_S & [\vec{e}_2]_S \end{bmatrix}$$

Note that $\vec{e}_1$ is the first vector in S as well so $[\vec{e}_1]_S = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

Write $[\vec{e}_2]_S = \begin{bmatrix} a \\ b \end{bmatrix}$, then $\vec{e}_2 = a\begin{bmatrix} 1 \\ 0 \end{bmatrix} + b\begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} a-b \\ 2b \end{bmatrix}$

$$\text{so } \begin{cases} a-b=0 \\ 2b=1 \end{cases} \Rightarrow \begin{cases} a=\frac{1}{2} \\ b=\frac{1}{2} \end{cases}, \quad \text{so } [\vec{e}_2]_S = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}. \quad \text{Thus } P_{S \leftarrow E} = \begin{bmatrix} 1 & \frac{1}{2} \\ 0 & \frac{1}{2} \end{bmatrix}$$

(c) Find the matrix representation of L with respect to the ordered basis S .

First let's find

$$P_{E \leftarrow S} = (P_{S \leftarrow E})^{-1}$$

(One can directly write down $P_{E \leftarrow S}$ just as what we did in part (b) as well.)

$$\left[\begin{array}{cc|cc} 1 & \frac{1}{2} & 1 & 0 \\ 0 & \frac{1}{2} & 0 & 1 \end{array} \right] \xrightarrow{-x_2 + x_1} \left[\begin{array}{cc|cc} 1 & 0 & 1 & -1 \\ 0 & \frac{1}{2} & 0 & 1 \end{array} \right] \xrightarrow{2x_2 \rightarrow x_2} \left[\begin{array}{cc|cc} 1 & 0 & 1 & -1 \\ 0 & \frac{1}{2} & 0 & 1 \end{array} \right]$$

$$\uparrow P_{E \leftarrow S} = P_{S \leftarrow E}^{-1}$$

Denote the matrix we got in (a) as A

and the matrix representation of L w.r.t. S as B then

$$B = P_{S \leftarrow E} A P_{E \leftarrow S} = \begin{bmatrix} 1 & \frac{1}{2} \\ 0 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & \frac{1}{2} \\ 0 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 1 & 5 \end{bmatrix} = \begin{bmatrix} \frac{5}{2} & \frac{5}{2} \\ \frac{1}{2} & \frac{5}{2} \end{bmatrix}$$

(You could also find B via $[L(\begin{bmatrix} 1 \\ 0 \end{bmatrix})]_S, [L(\begin{bmatrix} -1 \\ 2 \end{bmatrix})]_S$ or you can check answer this way.)

12. (10 points) Let $L : P_2 \rightarrow P_2$ be defined by

$$L(at^2 + bt + c) = (a + (1 - \lambda)b)t^2 + (b + c)t + (a + (1 - \lambda^2)c)$$

(a) Represent L by a matrix with respect to the standard basis $(t^2, t, 1)$ (Hint: Only λ should appear in the matrix as a variable.)

$$\begin{array}{l|l|l} L(t^2) = t^2 + 1 & \text{let } S = (t^2, t, 1), \text{ then} & \text{So the matrix } A \\ \text{plug in } \begin{cases} a=1 \\ b=0 \\ c=0 \end{cases} & [L(t^2)]_S = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} & \text{representing } L \text{ w.r.t. } S \\ & [L(t)]_S = \begin{bmatrix} 1-\lambda \\ 1 \\ 0 \end{bmatrix} & \text{is given by} \\ L(t) = (1-\lambda)t^2 + t & & \\ \text{plug in } \begin{cases} a=0 \\ b=1 \\ c=0 \end{cases} & [L(1)]_S = \begin{bmatrix} 0 \\ 1 \\ 1-\lambda^2 \end{bmatrix} & \\ L(1) = t + (1-\lambda^2) & & A = \begin{bmatrix} 1 & (1-\lambda) & 0 \\ 0 & 1 & 1 \\ 1 & 0 & (1-\lambda^2) \end{bmatrix} \end{array}$$

(b) For which values of λ the transformation is invertible?

$$L \text{ is invertible} \Leftrightarrow A \text{ is invertible} \Leftrightarrow \det A \neq 0.$$

$$\text{Note that } \det A = 1 \cdot (1 - \lambda^2) + 1 \cdot (1 - \lambda) = (1 + \lambda)(1 - \lambda) + (1 - \lambda) = (2 + \lambda)(1 - \lambda)$$

$$\text{So } \det A \neq 0 \Leftrightarrow \lambda \neq -2, 1.$$

Thus L is invertible for any λ NOT equal to $-2, 1$.

(c) When $\lambda = 1$, find all polynomials $f \in P_2$ such that $L(f) = 2t^2 + 3t + 2$.

$$\text{let } f = at^2 + bt + c.$$

$$\text{When } \lambda = 1, A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}.$$

We have

$$L(f) = 2t^2 + 3t + 2$$

$$\Leftrightarrow [L(f)]_S = \begin{bmatrix} 2 \\ 3 \\ 2 \end{bmatrix} \quad (\text{since } S = (t^2, t, 1))$$

$$\Leftrightarrow A[f]_S = \begin{bmatrix} 2 \\ 3 \\ 2 \end{bmatrix} \quad ([L(f)]_S = A[f]_S \text{ since } A \text{ represents } L \text{ w.r.t. } S)$$

$$\Leftrightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 2 \end{bmatrix} \Leftrightarrow \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 2 \\ -r+3 \\ r \end{bmatrix}, r \in \mathbb{R}.$$

$$\begin{bmatrix} 1 & 0 & 0 & | & 2 \\ 0 & 1 & 0 & | & 3 \\ 1 & 0 & 0 & | & 2 \end{bmatrix} \xrightarrow{-r_1 + r_3} \begin{bmatrix} 0 & 0 & 0 & | & 0 \\ 0 & 1 & 0 & | & 3 \\ 1 & 0 & 0 & | & 2 \end{bmatrix}$$

(continued on next page.)

Scratch Paper

So all the polynomials f satisfying $L(f) = 2t^2 + 3t + 2$ are those of the form:

$$f(t) = 2t^2 + (r+3)t + r \quad \text{for some } r \in \mathbb{R}.$$

↑
fixed my typo